

JAGAN BABU R.

Aspiring Software/ML Engineer

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Professional Summary

Aspiring Software/ML Engineer with hands-on experience building production-style AI systems and real-time ML inference. Interested in backend and applied AI roles.

Technical Skills

Languages & Databases: Python, Java, C++, C, SQL, PostgreSQL, MongoDB, Node.js

Backend & Tools: FastAPI, Flask, REST APIs, Docker, Linux (Ubuntu/Kali), Git, Cloud-Native Methodologies

Data Structures & Algorithms: Arrays, Strings, Stacks, Queues, Linked Lists, Trees, Graphs, Dynamic Programming, Sorting & Searching (400+ problems solved on Skillrack)

AI & ML: Generative AI, MLOps, LLMs/RAG, Computer Vision (CNNs), TensorFlow, Scikit-learn, Agentic Workflows

Automation & Systems: Core Systems Automation, ROS, PLC/SCADA Logic, Industrial Automation, System Design

Education

B.E. in Mechanical and Automation Engineering (CGPA: 8.57/10)

2023 – 2027

Sri Sairam Engineering College, Chennai, India

Experience

Intern, Electrical Maintenance / Rolling Stock

June 2025 – July 2025

Chennai Metro Rail Limited (CMRL), Chennai, India

- Optimized system reliability by analyzing SCADA and TCMS data streams across 15+ train sets, contributing to a 99.9% service availability benchmark through proactive software-driven fault triangulation
- Automated error logging mechanisms by reverse-engineering and analyzing PLC logic sequences for critical hardware systems, reducing diagnostic turnaround time by 20%

Projects

UPI Fraud Detection System | Python, XGBoost, GNN, LSTM, Isolation Forest | [Live Demo](#)

- Designed a scalable network-graph data processing pipeline handling 3.2M transaction nodes, leveraging advanced data structures and ensemble modeling to achieve a 0.9996 ROC-AUC with strict low-latency processing constraints
- Developed graph-based fraud detection with GNN and LSTM behavioral analysis, identifying 16,382 fraud nodes with a low false positive rate of 0.005

Cecilia AI (Agentic Workflow Assistant) | Python, FastAPI, RAG, LLMs

- Architected a high-performance backend workflow automation engine utilizing Python and FastAPI, implementing optimal data retrieval strategies to achieve sub-500ms response times and reduce manual intervention by 30-40%

VeriChain | Solidity, Node.js, Blockchain | [Live Demo](#)

- Developed a secure, decentralized digital credentialing infrastructure using Node.js and Solidity; implemented Soulbound Tokens (SBTs) to ensure data integrity, security, and instantaneous on-chain verification

Smart Waste Segregation System | CNN, TensorFlow, ROS, Mechanical Design

- Led the development of a CNN-based automated classification model with 90%+ detection accuracy
- Designed and sketched mechanical mechanisms for automated bin physical operation including waste disposal paths

Publications & Achievements

- Lead Author** – Conference paper on IoT-enabled smart waste segregation, IEOM 2025, VIT Vellore
- Top 5 (Finalist)** – VIT-AP LaunchPad Contest (out of 900+ teams nationwide)
- Top 6 (Finalist)** – Mastek DeepBlue Competition (out of 130+ teams nationwide)
- Top 100 (Regional)** – Google Solution Challenge 2024
- Semi-Finalist** – Empresario, IIT Kharagpur E-Cell